



MIMS FastTrack (FT) Web Service Guide

1. How to Integrate MIMS Database using FT Web Service:

a. Using FT Web Service:

- i. FT Web Service has been developed using above Java API library and is deployed in local client application server using “WAR”, Tomcat and JDK.
- ii. This provides better compatibility for all third-party clinical applications.
- iii. We offer both SOAP and REST Web Service options.
- iv. This FT Web Service installation would be done by MIMS API Support team.
- v. Compatible with both Windows and Linux environments

b. Database:

- i. MIMS Database is a proprietary database (like FastTrackData.mr2) and does not require any third-party database like SQL or Oracle to host it into your local environments.
- ii. MIMS Database would be configured in a specific path like (D:\MIMSDatabase\FastTrackData.mr2) and thereafter it would be refreshed every quarter by downloading the quarterly database release which would be done by your IT team.
- iii. MIMS Database would be hosted on the same server where your application is hosted.
- iv. Compatible with both Windows and Linux environments

c. API Call:

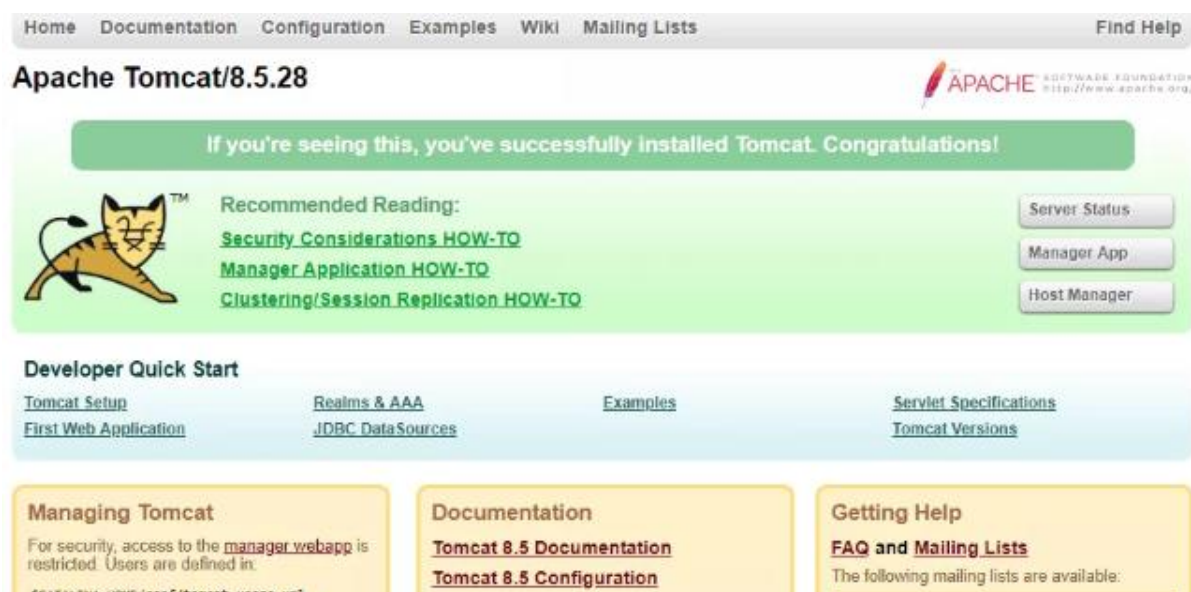
- i. Both Request API call and Result response would be in XML format which can be transformed into HTML to display on your application UI

2. FT Web Service Pre-requisites

- a. JDK Installation (preferably 1.8)
- b. Tomcat Server Installation (Version 9)

3. Testing Tomcat Installation

- Go to '<http://localhost:9090>', in case if 9090 is the port number of your tomcat.
- You should see the below page. It means you successfully installed Tomcat Server, else check if Tomcat has started if not you would need to troubleshoot.



4. FT Web Service Installation:

- RESTFTWebService.war:
 - Download the "RESTFTWebService.war" from MIMS FTP site (raise a request to MIMS)
 - Copy "RESTFTWebService.war" to ""C:\Program Files\Apache Software Foundation\Tomcat 8.5\webapps"

This PC > Local Disk (C:) > Program Files > Apache Software Foundation > Tomcat 9.0 > webapps			
Name	Date modified	Type	Size
docs	09-05-2019 18:40	File folder	
manager	09-05-2019 18:40	File folder	
RESTFTWebService	03-12-2019 11:09	File folder	
ROOT	09-05-2019 18:40	File folder	
RESTFTWebService.war	02-12-2019 10:23	WAR File	10,407 KB



b. "ft.properties" and "FastTrackData.mr2"

- i. Download the "**ft.properties**" file from MIMS FTP site (raise a request to MIMS)
- ii. Copy "**ft.properties**" to "**Database Path:**" (which was shared earlier via email to MIMS); example
- iii. Use the same step to refresh quarterly database file in the same path. But before replacing the file take a back and then refresh. Test this in UAT first, if successful then on PROD

Name	Date modified	Type	Size
MIMSstylesheet	03-12-2019 19:51	File folder	
FastTrackData.mr2	26-04-2019 04:16	MR2 File	6,83,249 KB
ft.properties	03-12-2019 19:52	PROPERTIES File	1 KB

c. Configuring "log4j" Path:

- i. Create a folder "**MIMSlog**" under "**Database Path:**" (which was shared earlier via email to MIMS) as shown below
- ii. This means the default path would be set to "C:\MIMSDatabase\MIMSlog"

Name	Date modified	Type	Size
MIMSlog	03-12-2019 20:03	File folder	
MIMSstylesheet	03-12-2019 19:51	File folder	
FastTrackData.mr2	26-04-2019 04:16	MR2 File	6,83,249 KB
ft.properties	03-12-2019 19:52	PROPERTIES File	1 KB

d. FT Web Service Link: (How to consume)

- i. Given below is a sample link which you can use to consume the service
- ii. IP Address and Port would point to your local server where FT Web Service is installed.

"xmlrequestplus" = <http://XXX.XX.XXX.XX:8081/RESTFTWebService/FTRequest/xmlrequestplus>

Port Number of your
application server

IP Address of your
application server

5. FT Web Service Methods:

Methods

- a. **"xmlrequestplus"** = ("prescriptionquery", "alertfilterbysverity", "alertfilterbydrug" or "responsetype")

Parameter List for API Call (Definition):

Mandatory Parameters:

- i. **"prescriptionquery"** = this corresponds to the request xml which you build to make the API call to MIMS Database.
- ii. **"responsetype"** = use this parameter to print the output as "xml" or "html"

Optional Parameters:

- i. **"alertfilterbyseverity"** = you can use this parameter to filter alerts by specific severity, else set this as "no" by default
- ii. **"alertfilterbydrug"** = you can use this parameter to filter alerts by drug GUID(single or multiple), else set this as "no" by default

Parameters

- a. **"prescriptionquery"** Parameter = FT Request.xml API call

Sample API call:

```
<Request>
<Interaction>
<Prescribing>
<Product reference="{C88D01B2-7969-49DB-A87B-6A370E5FB4FF}" />
<Product reference="{7211E99E-8A09-40CB-A20F-8FD840BEC083}" />
<Product reference="{3576E66D-1B46-47B4-B763-B9B34F9DE667}" />
<Product reference="{378D84FB-B50A-4B88-B537-D9AD346C7D0C}" />
<Product reference="{676A39F3-4D93-4A57-A920-CB551B0EA79A}" />
</Prescribing>
<References/>
</Interaction>
<PatientProfile>
<Gender>F</Gender>
<Pregnancy>
<Month>3</Month>
</Pregnancy>
<Age>
<Year>26</Year>
</Age>
<Nursing>true</Nursing>
</PatientProfile>
</Request>
```



- b. **“responsetype” Parameter** = “html” or “xml”
- c. **“alertfilterbydrug” Parameter** = use GUID to filter alerts for each or multiple drugs in a prescription

This feature supports filtering alerts for both single or multiple drugs using its GUID, see example below

Single Drug Example

```
<GUIDS>
  <GUID>{5FA1DA45-AD91-4BBA-A46C-7C16048D143D}</GUID>
</GUIDS>
```

Multiple Drug Example

```
<GUIDS>
  <GUID>{5FA1DA45-AD91-4BBA-A46C-7C16048D143D}</GUID>
  <GUID>{C9376526-AE03-4749-A71B-CF765155E14E}</GUID>
  <GUID>{B4374321-E067-1D25-E034-080020E1DD8C}</GUID>
  <GUID>{268C7977-533F-4765-A529-5AF2974F9F0D}</GUID>
  <GUID>{6924264D-5205-49EE-842E-B706B01F9207}</GUID>
  <GUID>{707BFAFF-1B76-4621-9983-D8212D2E00F7}</GUID>
  <GUID>{D52D4344-814B-44C8-9DA5-DE573964472C}</GUID>
</GUIDS>
```

- d. **“alertfilterbyseverity” Parameter** = use WarningID to display alerts with a list of preferred severities

This feature supports filtering alerts for using its Severity or WarningID, see example below

Query Syntax

```
<WARNINGIDS>
  <WARNINGID>D2D:Severe</WARNINGID>
  <WARNINGID>D2D:Moderate</WARNINGID>
  <WARNINGID>D2D:Minor</WARNINGID>
  <WARNINGID>D2D:Caution</WARNINGID>
  <WARNINGID>D2H:Contraindicated</WARNINGID>
  <WARNINGID>D2H:Extreme Caution</WARNINGID>
  <WARNINGID>DI:1</WARNINGID>
  <WARNINGID>DI:2</WARNINGID>
  <WARNINGID>DI:3</WARNINGID>
  <WARNINGID>DT:1</WARNINGID>
  <WARNINGID>DT:2</WARNINGID>
  <WARNINGID>DT:3</WARNINGID>
</WARNINGIDS>
```

6. How to Build XML request API Call:

Hospital Formulary Master

Sample Mapped File:

Hospital's Item code	Hospital's Item name (Brand name)	MIMS DESC	MIMS GUID	MIMS TYPE
DurgCode1	SIMARC 2MG TAB	warfarin 2mg Oral Tablet	{C26C4DD1-7BB2-4CF7-9329-B3CFC1DBAC29}	GGPI
DurgCode2	ABBOTIC XL 500MG TAB	clarithromycin 500mg Oral Tablet	{48FAE690-3728-4A4B-BA04-90044160B80B}	GGPI
DurgCode3	LIPITOR 20MG TAB	Lipitor FC tab 20 mg	{0C8DA0BB-4E41-4274-927C-D0C2340BC7FE}	Product
DurgCode4	FLUCONAZOLE 150MG TAB	fluconazole 150mg Oral Tablet	{2F83FC84-9BB8-4D54-A87B-FBD925691FA1}	GGPI
DurgCode5	PRO TB 4 TAB	ethambutol hydrochloride - isoniazid - pyrazinamide - rifampicin Oral Tablet	{EB883313-CB1A-4E33-93CE-C4C8A171124A}	GenericItem
DurgCode6	TEGRETOL 200MG TAB	Tegretol tab 200 mg	{D660DE2D-44B0-492F-98DE-9A6992D246AB}	Product
DurgCode7	CLOPIDOGREL 75MG TAB	clopidogrel 75mg Oral Tablet	{850620BD-554E-4D31-9DDB-8BD6A444C1A7}	GGPI
DurgCode8	PLAVIX 75MG TAB	Plavix FC tab 75 mg	{67DD8810-FC34-4253-99EF-96B33AEA9B7B}	Product
DurgCode9	AMIODARONE HCL 50MG/ML-3ML	amiodarone hydrochloride 150mg per 3mL Injection	{B45F1C0B-BD66-2784-E034-080020E1DD8C}	GGPI
DurgCode10	CORDARONE 150MG/3ML INJ	Cordarone inj 150 mg/3 mL	{7211E99E-8A09-40CB-A20F-8FD840BEC083}	Product

Hospital Allergy Master

Sample Mapped File

Hospital's Allergy Code	Hospital's Allergy Desc	MIMS DESC	MIMS GUID	MIMS TYPE
AllergyCode1	atorvastatin	atorvastatin	{DAC45C7E-0C5E-42B5-9FB5-3527A1230370}	Molecule
AllergyCode2	HMG-CoA reductase inhibitors	HMG-CoA reductase inhibitors	{A1907026-49A8-4760-A9D1-BBC83F71212F}	SubstanceClass
AllergyCode3	erythromycin	erythromycin	{4EB78A92-119E-4DE5-8F43-15548663ABB8}	Molecule
AllergyCode4	Amiodarone	Amiodarone	{722228FC-BBFF-473B-AD28-FF84B173CBDA}	SubstanceClass
AllergyCode5	warfarin	Warfarin	{4474F71F-26B0-411B-A3A2-16537132387F}	Molecule



Building Request.xml:

Lets consider below scenario of Patient A

Medicine Being Prescribed:

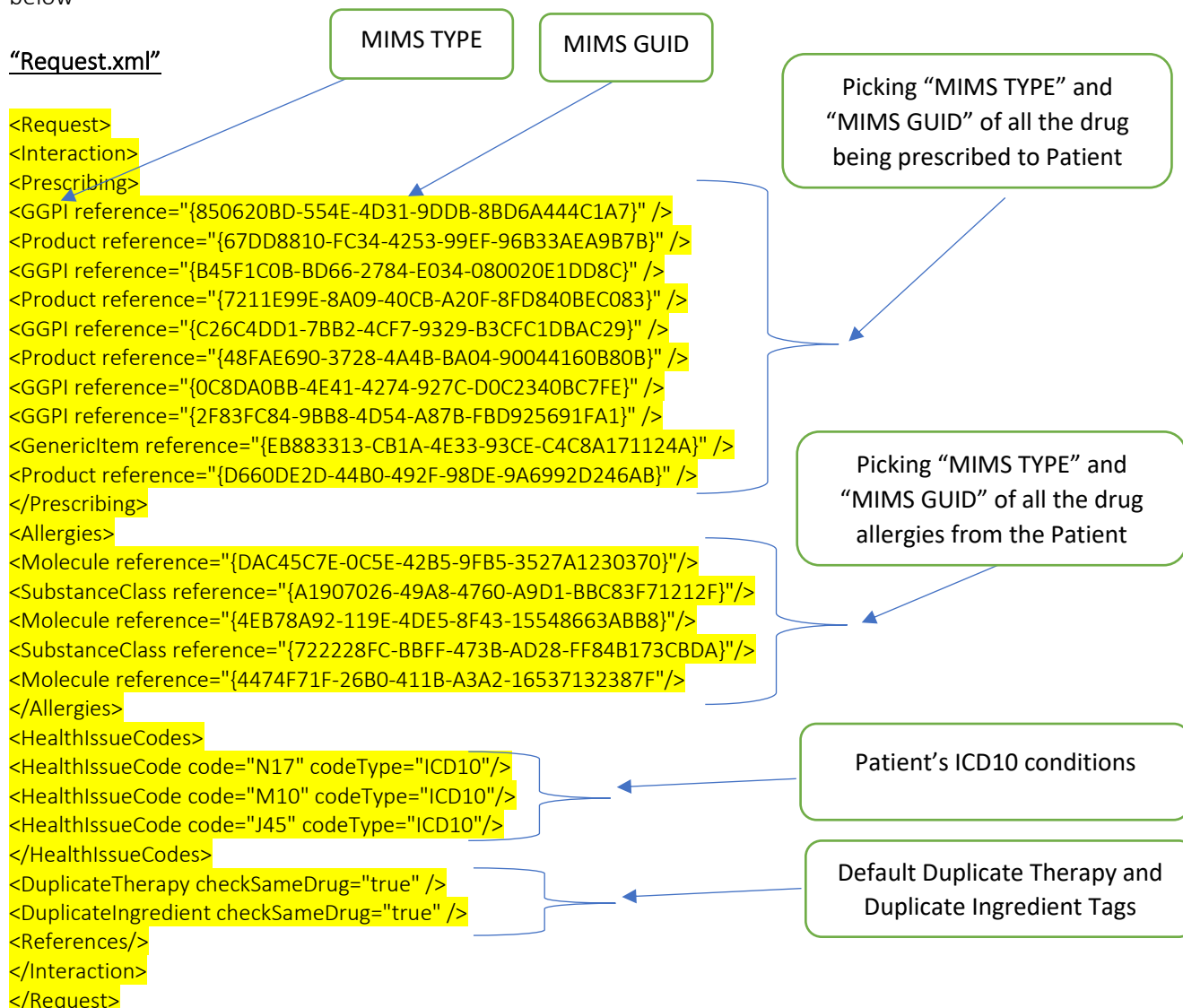
Hospital's Item code	Hospital's Item name (Brand name)
DurgCode1	SIMARC 2MG TAB
DurgCode2	ABBOTIC XL 500MG TAB
DurgCode3	LIPITOR 20MG TAB
DurgCode4	FLUCONAZOLE 150MG TAB
DurgCode5	PRO TB 4 TAB
DurgCode6	TEGRETOL 200MG TAB
DurgCode7	CLOPIDOGREL 75MG TAB
DurgCode8	PLAVIX 75MG TAB
DurgCode9	AMIODARONE HCL 50MG/ML-3ML
DurgCode10	CORDARONE 150MG/3ML INJ

Patient's existing Allergy Conditions:

Hospital's Allergy Code	Hospital's Allergy Desc
AllergyCode1	Atorvastatin
AllergyCode2	HMG-CoA reductase inhibitors
AllergyCode3	Erythromycin
AllergyCode4	Amiodarone
AllergyCode5	Warfarin



Now you pick the corresponding “MIMS TYPE” and “MIMS GUID” and build the “request.xml” as below



7. Client Source Code for Consuming above FT Web Service Methods:

```

import java.io.BufferedWriter;
import java.io.FileWriter;
import java.net.URI;
import java.util.HashMap;
import java.util.Iterator;
import java.util.Map;
import javax.ws.rs.core.UriBuilder;
import com.sun.jersey.api.client.Client;
  
```




```

import com.sun.jersey.api.client.WebResource;
import com.sun.jersey.api.client.config.ClientConfig;
import com.sun.jersey.api.client.config.DefaultClientConfig;
import com.sun.jersey.api.representation.Form;

public class xmlrequestplus_samplecode {

    public static void main(String[] args) throws InterruptedException {
        ClientConfig config = new DefaultClientConfig();
        Client client = Client.create(config);
        WebResource service = client.resource(getBaseURI());

        // "prescriptionquery" is a mandatory parameter to pass all valid request.xml api calls, see example below

        String prescriptionquery = "<Request>\r\n" +
            "<Interaction>\r\n" +
            "<Prescribing>\r\n" +
            "<GGPI reference=\"{488F9F61-5D37-4989-925E-1742FFFDAA9E}\"/>\r\n" +
            "<GGPI reference=\"{B3E6B75E-9519-6AE7-E034-080020E1DD8C}\"/>\r\n" +
            "<GGPI reference=\"{BF33752F-6062-0589-E034-0003BA299378}\"/>\r\n" +
            "<GGPI reference=\"{B3E696CF-71F8-6518-E034-080020E1DD8C}\"/>\r\n" +
            "<GGPI reference=\"{AA0D6353-F6E1-4AB1-AF1E-5E5F109A0706}\"/>\r\n" +
            "<GGPI reference=\"{B7AD6ED9-C22D-561C-E034-080020E1DD8C}\"/>\r\n" +
            "</Prescribing>\r\n" +
            "<Allergies>\r\n" +
            "<Molecule reference=\"{5143B2B1-DBED-45A2-8308-61636C3A2F0B}\"/>\r\n" +
            "<Molecule reference=\"{7142A19F-EDE2-4A7C-A30B-06AB60B6B71F}\"/>\r\n" +
            "<Molecule reference=\"{74DCDDAF-9349-430F-A2F2-B7F2BC5CD8AA}\"/>\r\n" +
            "<SubstanceClass reference=\"{8B9ECB52-CDC9-4460-AB40-6E3C7DF38B1E}\"/>\r\n" +
            "<SubstanceClass reference=\"{9E775009-231D-728D-E034-080020E1DD8C}\"/>\r\n" +
            "</Allergies>\r\n" +
            "<HealthIssueCodes>\r\n" +
            "<HealthIssueCode code=\"J45\" codeType=\"ICD10\"/>\r\n" +
            "<HealthIssueCode code=\"N17\" codeType=\"ICD10\"/>\r\n" +
            "<HealthIssueCode code=\"I49.5\" codeType=\"ICD10\"/>\r\n" +
            "</HealthIssueCodes>\r\n" +
            "<DuplicateTherapy checkSameDrug=\"true\"/>\r\n" +
            "<DuplicateIngredient checkSameDrug=\"true\"/>\r\n" +
            "<References/>\r\n" +
            "</Interaction>\r\n" +
            "</Request>" +
            """,

        // "responsetype" is a mandatory parameter which allows you to choose the output as "xml" or "html"
        String responsetype = "html";

        // "alertfilterbydrug" is an optional parameter to filter alerts by specific drug (single or multiple drugs), hence accordingly
        // include the GUID's of all those drugs whose alerts need to be displayed in the final alert response.

        // StringBuilder alertfilterbydrug = new StringBuilder(); // For GUID's
        // alertfilterbydrug.append("<GUIDS>"); //
        // alertfilterbydrug.append("<GUID>{B4268321-1B08-73AE-E034-080020E1DD8C}</GUID>");
        // alertfilterbydrug.append("</GUIDS>");

        // "alertfilterbyseverity" is an optional parameter to filter alerts by severity, hence include all severities to display alerts
        // only with these severities while alerts with other severities will be removed or not shown. Hence uncomment below 7 lines
        // to use this parameter.

        // StringBuilder alertfilterbyseverity = new StringBuilder();

```



```
// alertfilterbyseverity.append("<WARNINGIDS>");
// alertfilterbyseverity.append("<WARNINGID>D2D:Severe</WARNINGID>");
// alertfilterbyseverity.append("<WARNINGID>D2D:Moderate</WARNINGID>");
// alertfilterbyseverity.append("<WARNINGID>D2D:Minor</WARNINGID>");
// alertfilterbyseverity.append("<WARNINGID>D2D:Caution</WARNINGID>");
// alertfilterbyseverity.append("</WARNINGIDS>");
```

```
// "prescriptionquery" and "responsetype" are mandatory paramerters
Map<String, String> formDataMap = new HashMap<String, String>();
formDataMap.put("prescriptionquery", prescriptionquery.toString());
formDataMap.put("responsetype", responsetype);
```

```
// Both "alertfilterbydrug" and "alertfilterbyseverity" are optional paramerters, hence use or don't include this paramter
```

```
// formDataMap.put("alertfilterbydrug", alertfilterbydrug.toString());
// formDataMap.put("alertfilterbyseverity", alertfilterbyseverity.toString());
```

```
Form form = new Form();
Iterator<String> itr = formDataMap.keySet().iterator();
while (itr.hasNext()) {
    String key = itr.next();
    form.add(key, formDataMap.get(key));
}
```

```
// Below line will give you the filtered output
String response = service.path("FTRequest").path("xmlrequestplus").post(String.class, form);
System.out.println(response);
```

```
}
```

```
private static URI getBaseURI() {
    return UriBuilder.fromUri("http://localhost:9091/RESTFTWebService/").build();
}
```

```
}
```

8. Final Interaction Alert (“result.html”):

Drug Interaction (20)	Allergy (3)	Duplicate Therapy (4)	Duplicate Ingredient (2)
- amiodarone hydrochloride 150mg per 3mL Injection (amiodarone hydrochloride/Systemic) vs warfarin 2mg Oral Tablet (warfarin/Systemic) [Use combination with extreme caution.] - Cordarone inj 150 mg/3 mL (amiodarone hydrochloride/Systemic) vs warfarin 2mg Oral Tablet (warfarin/Systemic) [Use combination with extreme caution.] - amiodarone hydrochloride 150mg per 3mL Injection (amiodarone hydrochloride/Systemic) vs fluconazole 150mg Oral Tablet (fluconazole/Systemic) [Avoid combination.] - Cordarone inj 150 mg/3 mL (amiodarone hydrochloride/Systemic) vs fluconazole 150mg Oral Tablet (fluconazole/Systemic) [Avoid combination.] - warfarin 2mg Oral Tablet (warfarin/Systemic) vs ethambutol hydrochloride - isoniazid - pyrazinamide - rifampicin Oral Tablet (rifampicin/Systemic) [Use combination with caution.] - warfarin 2mg Oral Tablet (warfarin/Systemic) vs ethambutol hydrochloride - isoniazid - pyrazinamide - rifampicin Oral Tablet (isoniazid/Systemic) [Use combination with caution.] - warfarin 2mg Oral Tablet (warfarin/Systemic) vs fluconazole 150mg Oral Tablet (fluconazole/Systemic) [Use combination with caution.] - warfarin 2mg Oral Tablet (warfarin/Systemic) vs Tegretol tab 200 mg (carbamazepine/Systemic) [Use combination with caution.] - ethambutol hydrochloride - isoniazid - pyrazinamide - rifampicin Oral Tablet (isoniazid/Systemic) vs Tegretol tab 200 mg (carbamazepine/Systemic) [Use combination with caution.] - fluconazole 150mg Oral Tablet (fluconazole/Systemic) vs ethambutol hydrochloride - isoniazid - pyrazinamide - rifampicin Oral Tablet (rifampicin/Systemic) [Use combination with caution.] - fluconazole 150mg Oral Tablet (fluconazole/Systemic) vs Tegretol tab 200 mg (carbamazepine/Systemic) [Use combination with extreme caution.] - Plavix FC tab 75 mg (clopidogrel/Systemic) vs warfarin 2mg Oral Tablet (warfarin/Systemic) [Use combination with caution.] - clopidogrel 75mg Oral Tablet (clopidogrel/Systemic) vs warfarin 2mg Oral Tablet (warfarin/Systemic) [Use combination with caution.] - Plavix FC tab 75 mg (clopidogrel/Systemic) vs ethambutol hydrochloride - isoniazid - pyrazinamide - rifampicin Oral Tablet (rifampicin/Systemic) [Use combination with caution.] - clopidogrel 75mg Oral Tablet (clopidogrel/Systemic) vs ethambutol hydrochloride - isoniazid - pyrazinamide - rifampicin Oral Tablet (rifampicin/Systemic) [Use combination with caution.] - clopidogrel 75mg Oral Tablet (clopidogrel/Systemic) vs fluconazole 150mg Oral Tablet (fluconazole/Systemic) [Use combination with caution.] - Plavix FC tab 75 mg (clopidogrel/Systemic) vs fluconazole 150mg Oral Tablet (fluconazole/Systemic) [Use combination with caution.] - Cordarone inj 150 mg/3 mL (amiodarone hydrochloride/Systemic) vs ethambutol hydrochloride - isoniazid - pyrazinamide - rifampicin Oral Tablet (rifampicin/Systemic) [Use combination with caution.] - amiodarone hydrochloride 150mg per 3mL Injection (amiodarone hydrochloride/Systemic) vs ethambutol hydrochloride - isoniazid - pyrazinamide - rifampicin Oral Tablet (rifampicin/Systemic) [Use combination with caution.] - ethambutol hydrochloride - isoniazid - pyrazinamide - rifampicin Oral Tablet (rifampicin/Systemic) vs Tegretol tab 200 mg (carbamazepine/Systemic) [Use combination with caution.]			



9. Testing FT Web Service using POSTMAN

a. xmlrequestplus:

i. <http://XXX.XX.XXX.XX:8081/RESTFTWebService/FTRequest/xmlrequestplus>

POST <http://localhost:8080/RESTFTWebService/FTRequest/xmlrequestplus> [Send](#)

Params Authorization Headers (8) **Body** Pre-request Script Tests Settings [Cookies](#)

☐ none ☐ form-data ☒ x-www-form-urlencoded ☐ raw ☐ binary ☐ GraphQL

KEY	VALUE	DESCRIPTION	...	Bulk Edit
☒ prescriptionquery	<Request><Interaction><Prescribing><GGPI reference...			
☒ responsetype	xml			
☐ alertfilterbyseverity	no			
☐ alertfilterbydrug	<GUIDS><GUID>(1B464820-EC1B-460F-AF03-555567...			
Key	Value	Description		

Body Cookies Headers (5) Test Results [Status: 200 OK](#) [Time: 847 ms](#) [Size: 66.59 KB](#) [Save Response](#)

Pretty Raw Preview Visualize XML [≡](#)

```

1 <Result buildID="3493" copyright="Copyright (c) 2021 MIMS" expiryDate="2022-03-31">
2   <Interaction>
3     <GGPI reference="{BEBE223A-F917-046D-E034-0003BA299378}" name="adenosine 3mg per 1mL Injection" >
4       <Route name="Systemic" reference="{90075601-10D4-49DD-9292-98884D9D3EF3}">
5         <HealthIssueCode code="J45" codeType="ICD10" name="Asthma" >
6           <ClassInteraction>
7             <PrescribingInteractionClass name="Adenosine" reference="{CAE7CC54-0037-4F0F-AABB-C62F2F4D4C9C}" description="An endogenous
8               nucleoside with antiarrhythmic and vasodilator properties.">
9               <PrescribingMolecule name="adenosine" reference="{47150CD8-494D-4967-ADA7-F3F343865765}"/>
            </PrescribingInteractionClass>

```